

4. Project Planning Phase

Project Planning Phase

Introduction

Project planning is an essential phase that defines the roadmap for successful project completion. It includes task allocation, timeline estimation, milestone planning, and resource management. Proper planning ensures that every team member understands their responsibilities and that the project progresses in a systematic manner.

Project Objectives

- * Develop a Machine Learning model to predict Human Development Index (HDI).
- * Build a Flask-based web application for user interaction.
- * Perform data preprocessing and visualization.
- * Evaluate the model using regression metrics.
- * Deploy a user-friendly prediction system.

Team Members and Responsibilities

Team Member	Responsibility
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Nagella Sravani	Project Workflow, Environment Setup, Flask Integration, GitHub Management
Nagalli Deepak	Dataset Collection, Data Exploration, Data Visualization
Ugadi Shiva Kumar	Data Preprocessing, Feature Selection, Missing Value Handling
Nallagoti Lavanya	Train-Test Split, Prediction Module, Testing
Vannala Madhusekhar	Model Training, Model Saving, Flask Backend

Sprint Planning

Sprint 1

- * Install required libraries

- * Create project structure

- * Download dataset

Sprint 2

- * Data preprocessing

- * Data visualization

- * Feature selection

Sprint 3

- * Train Linear Regression model

- * Evaluate performance

- * Save model

Sprint 4

- * Develop Flask application

- * Design HTML pages

- * Perform testing

- * Upload to GitHub

Project Timeline

Week 1

- * Requirement Analysis

- * Brainstorming

Week 2

- * Dataset Collection

- * Data Exploration

Week 3

- * Model Development

Week 4

- * Flask Deployment

* Documentation

* Testing

Milestones

✓ Environment Setup

✓ Dataset Collection

✓ Data Preprocessing

✓ Model Training

✓ Model Evaluation

✓ Model Saving

✓ Flask Deployment

✓ Documentation

✓ Final Demonstration

Conclusion

The planning phase ensured efficient task distribution, proper time management, and successful completion of the project within the specified timeline.